COMPARATIVE ANALYSIS TO IDENTIFY MACHINE LEARNING METHODS IN PREDICTING BENIGN-MALIGNANT TYPES OF BREAST CANCER

Anak Agung Ngurah Gunawan[a*], Putu Astri Novianti[b], Anak Agung Ngurah Frady Cakra Negara[c], Anak Agung Ngurah Bagaskara[a], Nyoman Gunantara[d].

[a]*Department of Physics, Faculty of Mathematics and Natural Sciences, Udayana University, Bali, Indonesia*

[b]*General Surgery Department, Udayana University Hospital, Bali, Indonesia*

[c]*Department of Informatics, Faculty of Mathematics and Natural Sciences, Udayana University, Bali, Indonesia*

[d]*Department oftechnical electro,Udayana University, Bali, Indonesia*

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ABSTRACT

Breast cancer is a health problem in the world. This research aimed to comparatively analyze and identify machine learning methods in predicting benign-malignant types of breast cancer. This research was quantitative research. The dataset was taken by Kaggle Breast Cancer Wisconsin, berlicense CC BY-NC-SA 4.0, with URL address: https://www.kaggle.com/datasets/uciml/breast-cancer-wisconsin-data. There were 569 data, consisting of 212 malignant and 357 benign. This research used three types of data, namely original data, binary data transformation, and bipolar data transformation. Besides, this research also used six machine learning methods, namely Decision Tree, Naïve Bayes, Random Forest, Support Vector Machine (SVM), Artificial Neural Network (ANN), and K-Nearest Neighbor (KNN). Furthermore, this research compared the method with the highest TP, TN, accuracy, sensitivity, specificity, precision, and TPR values, and the method with the lowest FN and FP values. In addition, this research also compared which method had the widest ROC graph.Binary and bipolar transformations had the highest true positive values using the ANN method, followed by the SVM method. Binary and bipolar data transformations with the lowest false negatives using the ANN method, followed by the Random Forest and KNN methods. Binary and bipolar transformations had the highest accuracy using the KNN method. Binary and bipolar data transformations had the widest ROC graphs using the KNN method, followed by Random Forest and ANN. Based on the six machine learning methods analyzed, it was found that the KNN method was the most accurate method in predicting benign-malignant breast cancer. Data transformation results showed that almost all data transformations performed better than the original data.

1. Introduction

Cancer has been identified as a dangerous disease and the second cause of death worldwide [1]. Breast cancer mostly occurs among women and is one of the five causes of cancer-related deaths [2]. Breast cancer is one of the most common primary malignant tumors in women in China [3], [4]. Many women die from breast and breast disease. Most deaths from breast cancer occur due to metastasis [5], [6]. According to WHO, there were 2.3 million women diagnosed with breast cancer, with 685,000 deaths in 2020 [7], [8] Cancer Society Statistics reported that there were 281,550 new cases of invasive breast cancer diagnosed among American women and 43,600 deaths from breast cancer in 2021 [9]. In 2022, there were 1,918,030 new cancer cases and 609,360 deaths due to cancer in the United States [10]. In 2024, there were 2,001,140 new cancer cases and 611,720 deaths from cancer in the United States [11]. Considering the high incidence of breast cancer, early detection is essential. Early detection and diagnosis accuracy are necessary for the treatment and prognosis of breast cancer [12]. Mammography scans are commonly used as a source for early detection [13]. Since the 1980s, Mammography screening has become the cornerstone of early breast cancer detection [14]. Mammography can also improve the survival of breast cancer patients [15]. MRI has a high sensitivity for detecting breast lesions, especially in patients at high risk of breast cancer [16]. Early diagnosis can reduce breast cancer death rates [17]. Many researchers are competing to create new methods for early detection of breast cancer. Osman (2020) developed the Adjusted Quick Shift Phase Preserving Dynamic Range Compression method for breast cancer segmentation [18]. Koji Ohnuki (2021) developed a combination of mammography and ultrasound for breast screening [19].

Correspondence
Department of Physics, Faculty of Mathematics and Natural Sciences, Udayana University, Bali, Indonesia
E-mail: a.a.ngurahgunawan@unud.ac.id (A.A.N. Gunawan)

Lan-Anh Dang (2022) developed artificial intelligence with mammography for breast screening [20]. Jakob Olinder (2023) developed a combined digital breast tomosynthesis and digital mammography for breast screening [21]. Yedda Nunes Reis (2023) evaluated pathological macroscopic breast density with mammographic breast density in breast cancer conservation surgery [22]. Machine learning has been widely developed for early detection of breast cancer. Kimberlee A. Hashiba (2023) used Machine Learning Models for surgical risk prediction Upstaging Ductal Carcinoma In Situ [23]. Vincent Peter C. Magboo (2021) developed Machine Learning for a Breast Cancer Recurrence Classifier [24]. Wei-Chung Shia (2021) developed machine learning to classify malignant tumors on breast ultrasound [25]. Richard Adam1 (2023) developed machine learning for breast cancer detection with magnetic resonance imaging [26]. Marion Olubunmi Adebiyi (2022) developed machine learning to classify and predict benign or malignant cancer [27]. Researchers are starting to compete to improve machine learning performance. Agaba Ameh Joseph (2022) used hand-crafted features and deep neural networks to enhance the multi-classification of breast cancer histopathological images [28]. Wenlong Ming (2022) used a transfer learning technique to predict hormone receptor and PAM50 breast cancer subtypes from multi-scale lesion images of DCE-MRI [29]. Bita Asadi (2023) used a deep learning network to streamline breast cancer detection [30]. Ming Y. Lu (2022) used Deep Learning to characterize morphological phenotypes and predict features that humans cannot identify from histology, such as molecular changes [31]. Hanan Aljuaid (2022) used deep neural networks and transfer learning for breast cancer classification [32]. Many do not know that binary or bipolar data transformation can improve machine learning performance. Therefore, this research aimed to comparatively analyze and identify machine learning methods in predicting benign-malignant types of breast cancer. This research is fundamental to conduct, considering that the breast cancer death rate is increasing every year.

1. Methods

This research was quantitative research. The dataset was taken by Kaggle Breast Cancer Wisconsin, berlicense CC BY-NC-SA 4.0, with URL address: https://www.kaggle.com/datasets/uciml/breast-cancer-wisconsin-data. There were 569 data, consisting of 212 malignant and 357 benign. The data was divided randomly, consisting of 70% for training and 30% for testing. The dataset was in the form of features taken from the Fine Needle Aspirate (FNA) results of suspicious mass images. There were ten image features from each cell nucleus in the form of radius, texture, perimeter, area, smoothness, compactness, concavity, concave points, symmetry, and fractal dimension, as shown in Figure 2. Fr

om each image feature, the mean, standard error, and worst were calculated so that the total number of features was 30 features. This research used three types of data, namely original data, binary data transformation, and bipolar data transformation. Besides, this research also used six machine learning methods, namely Decision Tree, Naïve Bayes, Random Forest, Support Vector Machine (SVM), Artificial Neural Network (ANN), and K-Nearest Neighbor (KNN). Furthermore, this research compared the TP, FP, FN, TN, accuracy, sensitivity, specificity, precision, TPR, and FPR values, as shown in Table 1, Table 2, Figure 4 to Figure 16. From Table 1, Table 2, Figure 4, and Figure 7, this research could conclude which method had the highest accuracy and the lowest false negative values. Then, this research also analyzed the ROC graph to see the widest ROC graph of this research-proposed method, as shown in Figure 17. The research flowchart from this research is shown in Figure 1.

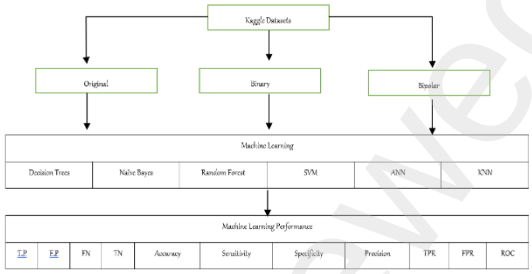

Figure 1. Research Flowchart

1. Results

Visualization of 10 data features as shown in Figure 1.

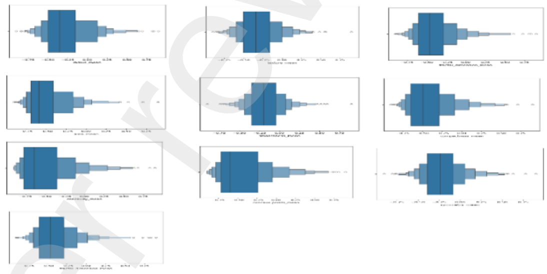

Figure 1. Visualization of 10 Data Features

Visual attributes of each as shown in Figure 2.

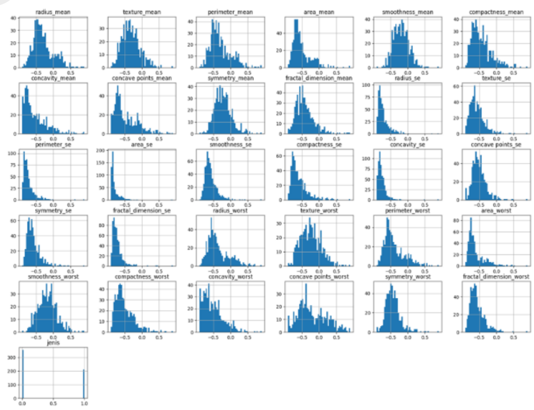

Figure 2. Visual Attributes of Each

Exploratory Data-Multivariate Analysis as shown in Figure 3.

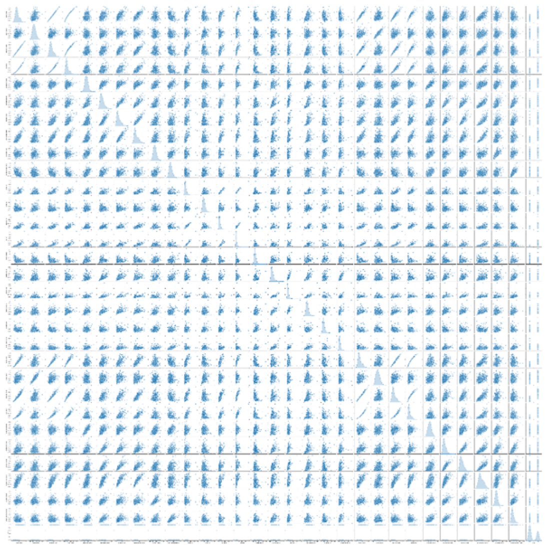

Figure 3. Exploratory Data-Multivariate Analysis

The calculation results of the TP, FP, FN, and TN values from each method can be seen in Table 1.
Table 1. TP, FP, FN, and TN Values from Each Method

		T.P	F.P	FN	TN
Decision Trees	Original	57	5	5	104
	Binary	56	7	11	97
	Bipolar	64	5	8	94
Naïve Bayes	Original	54	6	7	104
	Binary	57	9	6	99
	Bipolar	58	11	5	97
Random Forest	Original	57	4	4	106
	Binary	61	4	2	104
	Bipolar	62	5	1	103
SVM	Original	72	1	13	85
	Binary	78	0	7	86
	Bipolar	79	0	6	86
ANN	Original	83	21	2	65
	Binary	85	23	0	63
	Bipolar	85	21	0	65
KNN	Original	57	5	4	105
	Binary	66	2	3	100
	Bipolar	66	2	3	100

The calculation results of the Accuracy, Sensitivity, Specificity, and Precision values for each method can be seen in Table 2.
Table 2. Accuracy, Sensitivity, Specificity, and Precision Values of Each Method

		Accuracy (%)	Sensitivity (%)	Specificity (%)	Precision (%)	TPR	FPR
Decision Trees	Original	94.15	91.93	95.41	91.94	0.92	0.05
	Binary	89.47	83.58	93.27	88.89	0.83	0.07
	Bipolar	92.40	88.89	94.95	92.75	0.89	0.05
Naïve Bayes	Original	92.40	88.53	94.55	90	0.89	0.05
	Binary	91.23	90.48	91.67	86.36	0.91	0.08
	Bipolar	90.64	92.06	89.81	84.06	0.92	0.1
Random Forest	Original	95.32	93.44	96.36	93.44	0.93	0.04
	Binary	96.49	96.83	96.30	93.85	0.97	0.03
	Bipolar	96.49	98.41	95.37	92.54	0.98	0.05
SVM	Original	91.81	84.71	98.84	98.63	0.85	0.01
	Binary	95.91	91.76	100	100	0.92	0.0
	Bipolar	96.49	92.94	100	100	0.92	0.0
ANN	Original	86.55	97.65	75.58	79.81	0.98	0.24
	Binary	86.55	100	73.26	78.70	1.0	0.27
	Bipolar	87.72	100	75.58	80.19	1.0	0.24
KNN	Original	94.74	93.44	95.45	91.94	0.93	0.04
	Binary	97.08	95.65	98.04	97.06	0.96	0.02
	Bipolar	97.08	95.65	98.04	97.06	0.96	0.02

Based on the six methods analyzed, only four methods could increase the accuracy value by using data transformation, such as the Random Forest, SVM, ANN, and KNN methods, as shown in Figure 4.

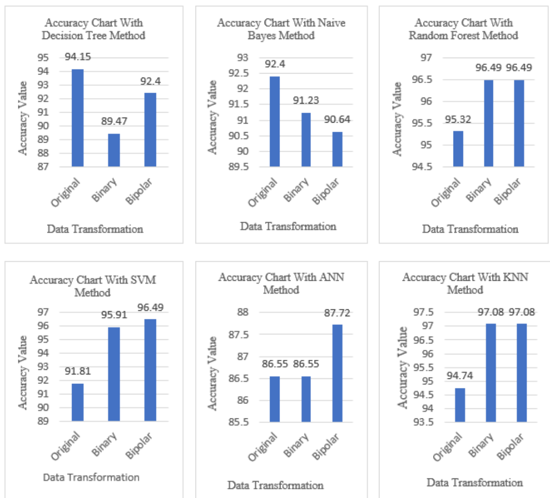

Figure 4. Accuracy Improvement Graph

Based on the six methods analyzed, data transformation could increase the true positive value, as shown in Figure 5

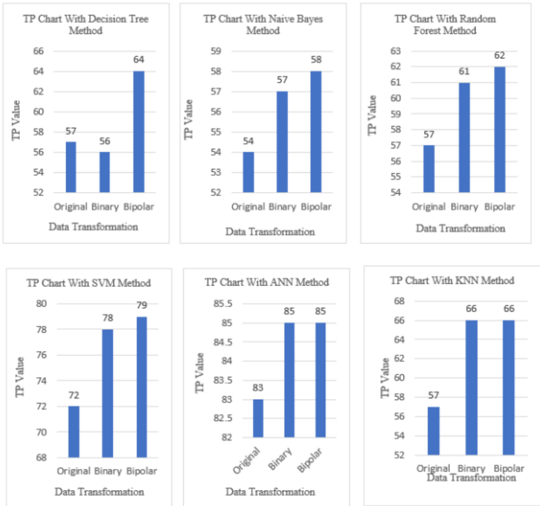

Figure 5. Graph of Increase in True Positive Value

Based on the six methods analyzed, only five methods could increase the true positive rate by using data transformation, such as the Naïve Bayes, Random Forest, SVM, ANN, and KNN methods, as shown in Figure 6.

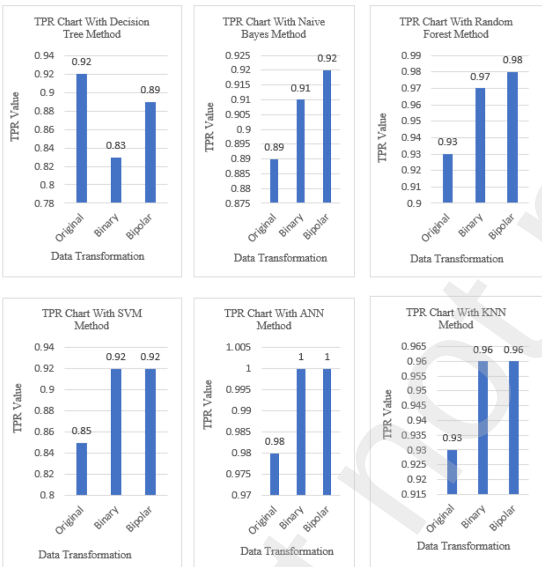

Figure 6. Graph of Increase in True Positive Rate Value

Based on the six methods analyzed, only five methods could reduce false negative values using data transformation, such as the Naïve Bayes, random forest, SVM, ANN, and KNN, as shown in Figure 7.

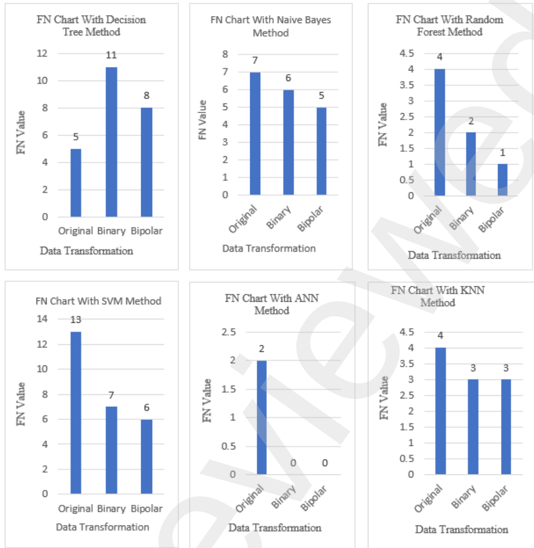

Figure 7. Graph of Decrease in False Negative Values

The decision tree method with bipolar transformation had the highest True Positive value, followed by the original type and binary transformation. The Naïve Bayes method with bipolar transformation had the highest True Positive value, followed by binary and original transformations. The Random Forest method with bipolar transformation had the highest True Positive value, followed by binary and original transformation. The SVM method with bipolar transformation had the highest True Positive value, followed by binary and original transformation. The ANN method with binary and bipolar transformations had equally high True Positive values, followed by the original type. The KNN method with binary and bipolar transformations had equally high True Positive values, followed by the original type, as shown in Figure 8.

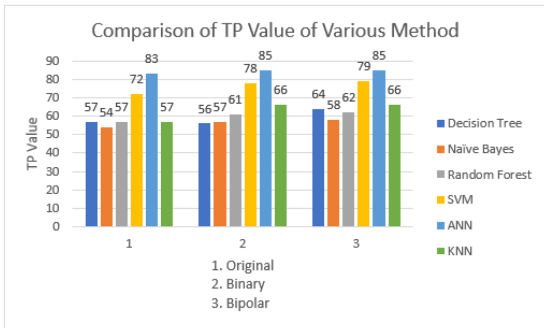

Figure 8. Comparison of TP Values of Each Method

The decision tree method with original type and bipolar transformation had equally low false positive values, followed by binary transformation. The Naïve Bayes method with the original type had the lowest false positive value, followed by binary and bipolar transformations. The Random Forest method with original type and binary transformation had equally low false positive values, followed by bipolar transformation. The SVM method with binary and bipolar transformations had equally low false positive values, followed by the original type. The ANN method with original type and bipolar transformation had equally low false positive values, followed by binary transformation. The KNN method with binary and bipolar transformations had equally low false positive values, followed by the original type, as shown in Figure 9.

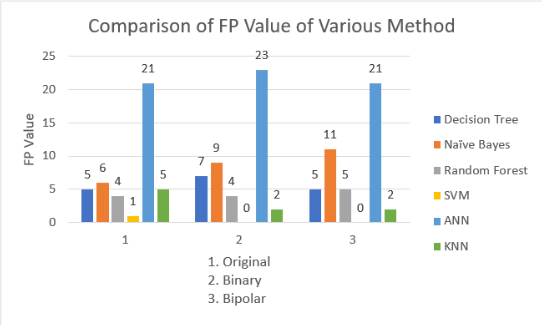

Figure 9. Comparison of FP Values of Each Method

The decision tree method with the original type had the lowest false negative value, followed by transformation bipolar and binary. The Naïve Bayes method with bipolar transformation had the lowest false negative value, followed by binary and original transformations. The Random Forest method with bipolar transformation had the lowest false negative value, followed by binary transformation and original type. The SVM method with bipolar transformation had the lowest false negative value, followed by binary transformation and original type. The ANN method with binary and bipolar transformations had equally low false negative values, followed by the original type. The KNN method with binary and bipolar transformations had equally low false negative values, followed by the original type, as shown in Figure 10.

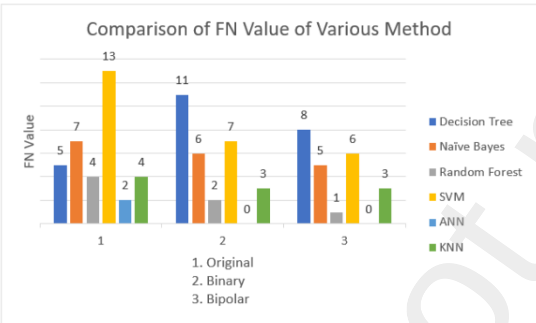

Figure 10. Comparison of FN Values of Each Method

The decision tree method with the original type had the highest True Negative value, followed by binary and bipolar transformations. The Naïve Bayes method with the original type had the highest True Negative value, followed by binary and bipolar transformations. The Random Forest method with the original type had the highest True Negative value, followed by binary and bipolar transformations. The SVM method with binary and bipolar transformations had the same True Negative value as the highest, followed by the original type. The ANN method with original type and bipolar transformation had equally high True Negative values, followed by binary transformation. The KNN method with the original type had the highest True Negative value, followed by binary and bipolar transformations, as shown in Figure 11.

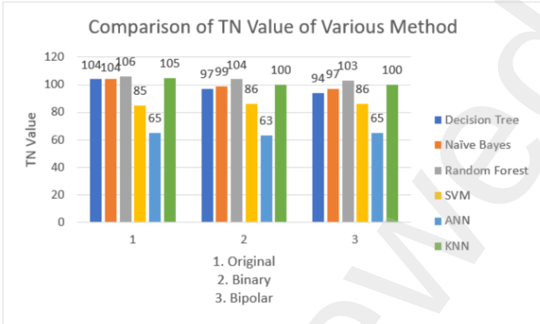

Figure 11. Comparison of TN Values of Each Method

The decision tree method with the original type had the highest accuracy value, followed by bipolar and binary transformations. The Naïve Bayes method with the original type had the highest accuracy value, followed by binary and bipolar transformations. The Random Forest method with binary and bipolar transformations had equally high accuracy values, followed by the original type. The SVM method with bipolar transformation had the highest accuracy value, followed by binary transformation and original type. The ANN method with bipolar transformation had the highest accuracy value, followed by binary transformation and original type. KNN method with transformation binary and bipolar had equally high accuracy values, followed by the original type, as shown in Figure 12.

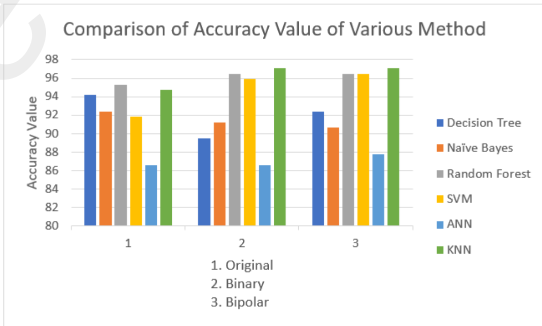

Figure 12. Comparison of Accuracy Values of Each Method

The decision tree method with the original type had the highest sensitivity value, followed by bipolar and binary transformations. The Naïve Bayes method with bipolar transformation had the highest sensitivity value, followed by binary transformation and original type. The Random Forest method with bipolar transformation had the highest sensitivity value, followed by binary transformation and original type. The SVM method with bipolar transformation had the highest sensitivity value, followed by binary transformation and original type. The ANN method with binary and bipolar transformations had equally high sensitivity values, followed by the original type. The KNN method using binary and bipolar transformations had equally high sensitivity values, followed by the original type, as shown in Figure 13.

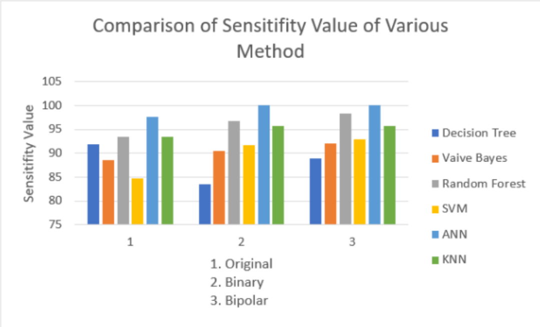

Figure 13. Comparison of Sensitivity Values of Each Method

The decision tree method with the original type had the highest Specificity value, followed by bipolar and binary transformations. The Naïve Bayes method with the original type had the highest Specificity value, followed by binary and bipolar transformations. The Random Forest method with the original type had the highest Specificity value, followed by binary and bipolar transformations. The SVM method with binary and bipolar transformations had equally high Specificity values, followed by the original type. The ANN method with original type and bipolar transformation had equally high Specificity values, followed by binary transformation. The KNN method using binary and bipolar transformations had equally high Specificity values, followed by the original type, as shown in Figure 14.

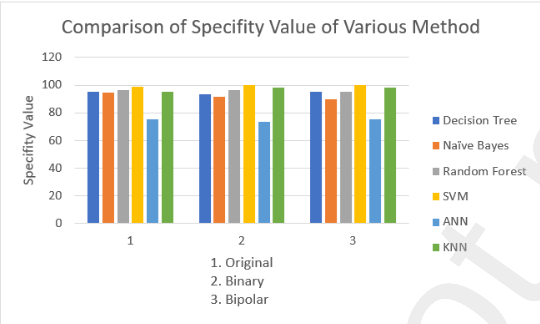

Figure 14. Comparison of Specificity Values of Each Method

The decision tree method with bipolar type had the highest precision value, followed by the original type and binary transformation. The Naïve Bayes method with the original type had the highest precision value, followed by binary and bipolar transformations. The Random Forest method with binary type had the highest precision value, followed by the original type and bipolar transformation. The SVM method with binary and bipolar transformations had equally high precision values, followed by the original type. The ANN method with bipolar type and bipolar transformation had equally high Precision values, followed by the original type and binary transformation. The KNN method using binary and bipolar transformations had equally high precision values, followed by the original type, precision as shown in Figure 15.

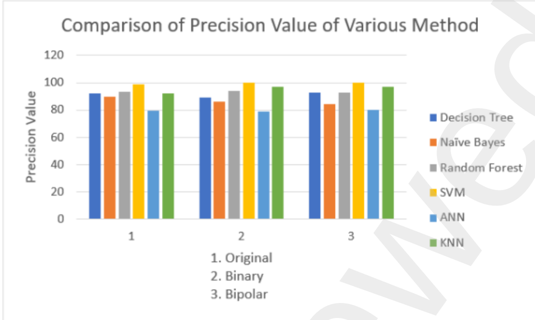

Figure 15. Comparison of Precision Values of Each Method

The decision tree method with the original type had the highest TPR value, followed by bipolar and binary transformations. The Naïve Bayes method with bipolar transformation had the highest TPR value, followed by binary and original transformation. The Random Forest method with bipolar transformation had the highest TPR value, followed by binary and original transformation. The SVM method with binary and bipolar transformations had equally high TPR values, followed by the original type. The ANN method with binary and bipolar transformations had equally high TPR values, followed by the original type. The KNN method using binary and bipolar transformations had equally high TPR values, followed by the original type, as shown in Figure 16.

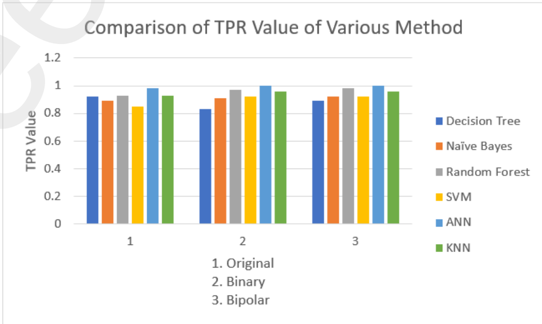

Figure 16. Comparison of TPR Values of Each Method

Based on the six methods analyzed using binary and bipolar data transformation, only ANN had the widest ROC graph, followed by random forest and KNN, as shown in Figure 17.

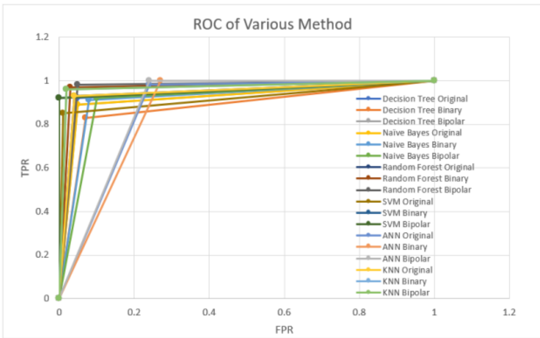

Figure 17. Comparison of ROC Graphs of Various Methods

Discussions

Binary and bipolar transformations had the highest true positive values with the ANN method, followed by the SVM method. Binary and bipolar data transformation using the SVM method had the lowest false positives, followed by the KNN method. Binary and bipolar data transformation using the ANN method had the lowest false negatives, followed by the Random Forest and KNN methods. Based on the three transformations proposed using six types of machine learning, the original type had the highest true negative value, followed by the binary and original transformations. Based on the six methods analyzed, the KNN method with binary and bipolar transformations had the highest accuracy. Based on the six methods analyzed, the ANN method with binary and bipolar transformation had the highest sensitivity, followed by the random forest method. The SVM method with binary and bipolar transformations had the highest specificity, followed by the KNN method. The SVM method with binary and bipolar transformations had the highest precision, followed by the KNN method. Based on the six methods analyzed, the ANN method with binary and bipolar transformations had the highest TPR, followed by the Random Forest method. The performance improvement of machine learning can be seen from the increase in True Positive value, accuracy, True Positive Rate, and decrease in false negative value. Besides the four types of variables above, this research also analyzed the ROC graph. The wider the ROC graph, the better the performance of machine learning. This research used three types of data transformation, namely original, binary, and bipolar data. This research compared these three types of data. It showed that the bipolar transformation type was the best, followed by binary and original transformations, as shown in Table 1, Table 2, and Figures 4 to Figure 8. Literature studies showed that the comparison of accuracy, sensitivity, and specificity values with sophisticated methods others can be seen in Table 3.

Table 3. Summary of Performance Results of Other Advanced Methods

Study	Year	Method	Accuracy	Sensitivity	Specificity
Bita Asadi et al. [30]	2023	Deep Learning Network ResNet50	0.986	0.986	Not mentioned
Hanan Aljuaid et al. [32]	2022	Deep Neural Networks and Transfer Learning	0.978	0.976	0.973
Saikat Islam Khan et al[33]	2022	Deep Neural Network MultiNet	0.99	Not mentioned	Not mentioned
Seifedine Kadry et al. [34]	2023	Pre-Trained Deep Learning Schemes	0.955	Not mentioned	Not mentioned
Sushovan Chaudhury et al. [17]	2023	Fast AI Technique and Sqeezenet Architecture Sushovan	0.903	Not mentioned	Not mentioned
Marco Caballo et al. [35]	2020	Deep learning-Based	0.92	Not mentioned	Not mentioned
Yuchao Zheng et al. [36]		Transfer Learning and Ensemble Learning	0.989	Not mentioned	Not mentioned
Zahra Assari et al. [37]	2022	Deep Residual Learning-Based	0.917	0.898	0.938
Yunan Wu et al. [38]	2022	Deep Learning Fusion Models	0.877	0.861	Not mentioned
Agaba Ameh Joseph et al. [28]	2022	Handcrafted Features and Deep Neural Networks	0.979	Not mentioned	Not mentioned
Imran Ul Haq et al. [13]	2022	Feature Fusion and Ensemble Learning-based CNN Models	0.994	0.995	0.994
Yew Sum Leong et al. [39]	2022	Deep Convolutional Neural Networks	0.999	0.999	1,000
Jeremy M. Webb et al. [40]	2021	Deep learning-Based	Not mentioned	0.826	0.993
Dan Song et al[41]	2022	Deep Multi-View Fusion	0.850	0.957	0.716
Md. Mahbubur Rahman et al. [8]	2022	Multi-Scale Feature Fusion	0.992	Not mentioned	Not mentioned
Wei-Chung Shia et al. [25]	2021	Pre-Trained Deep Residual Network Model and Support Vector Machine	Not mentioned	0.943	0.932
Xin Yu Liew et al. [42]	2021	XGBoost-Based Algorithm	0.97	Not mentioned	Not mentioned
Khanabhorn Kawattikul et al. [43]	2022	Conventional Handcrafted Feature and Deep-Learning Technique	Not mentioned	0.82	0.85
B.G. Deepa et al. [44]	2022	Convolutional Neural Network Models	0.945	Not mentioned	Not mentioned
Eroglu et al. [45]	2021	Convolutional Neural Networks	0.956	Not mentioned	Not mentioned
Nelsonet al. [46]	2023	Star-Convex Polygon	0.876	Not mentioned	Not mentioned
Abdolahi et al. [47]	2020	Artificial Intelligence	0.85	Not mentioned	Not mentioned
El Houby et al. [48]	2021	Convolutional Neural Networks	0.965	0.965	0.965
Osman et al. [18]	2020	Adjusted Quick Shift Phase Preserving Dynamic Range Compression	Not mentioned	0.795	0.979
Huang et al. [49]	2021	Convolutional Neural Networks (CNN), AlexNet, DenseNet, and ShuffleNet	0.955;	Not mentioned	Not mentioned

			0.997; 0.978		
Akinnuwesi et al. [50]	2020	Support Vector Machines	0.976	0.952	1.00
Chen et al. [51]	2023	Artificial Intelligence	0.778	0.921	0.671
Proposed	2024	Data Transformation	0.971	0.957	0.980

Conclusions

Machine learning had a significant role in predicting benign-malignant breast cancer. This research used six methods from machine learning to predict the type of benign-malignant breast cancer. This research observed the widest ROC graph, the highest accuracy value, and the lowest false negative value as the accuracy indicators of benign-malignant types. Based on the six machine learning methods analyzed, the KNN method was the most accurate method in predicting benign-malignant breast cancer. This research transformed the original data into binary and bipolar data in analyzing the benign-malignant type of breast cancer. Data transformation results showed that almost all transformed data performed better than the original data. This research contributed to providing knowledge about the use of data transformation to identify the best machine learning model in predicting benign-malignant breast cancer.

List of Abbreviations

SVM : Support Vector Machine
ANN : Artificial Neural Networks
KNN : K-Nearest Neighbor
T.P : Predicted to be malignant, actually malignant
TN : Predicted to be benign actually benign
FN : Predicted to be benign actually malignant
F.P : Predicted to be malignant but actually benign
ROC : Receiver Operating Characteristics
TPR : True Positive Rate
FPR : False Positive Rate

Competing interests

The authors declare that they have no competing interests

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Author Contributions

all authors have read and approved the manuscript
AANG: project design, data analysis, script writing; AANF: machine learning program; AANB: data analysis; PAN: Manuscript editing, data collection.

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